



Semester One Examination, 2025

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section Two: Calculator-assumed

SOLUTIONS

WA student number: In figures

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In words

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Your name

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Time allowed for this section

Reading time before commencing work: ten minutes

Working time: one hundred minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators, which can include scientific, graphic and Computer Algebra System (CAS) calculators, are permitted in this ATAR course examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	10	10	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed**65% (98 Marks)**

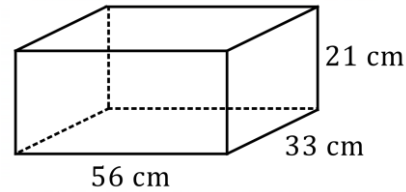
This section has **ten** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 8

(10 marks)

A closed rectangular cardboard box has a height of 21 centimetres and base dimensions of 56 by 33 centimetres as shown.



- (a) Determine the volume of the box.

(1 mark)

Solution
$V = 56 \times 33 \times 21 = 38\,808 \text{ cm}^3$
Specific behaviours
✓ correct volume

- (b) Determine the total surface area of the six faces of the box.

(3 marks)

Solution
$A = 2 \times (56 \times 33 + 56 \times 21 + 33 \times 21)$ $= 2 \times (1848 + 1176 + 693)$ $= 2 \times 3717 = 7434 \text{ cm}^2$
Specific behaviours
✓ indicates one correct rectangular area ✓ indicates all correct rectangular areas ✓ correct total surface area

- (c) The 12 edges of the box are to be reinforced with rubber tape. Determine the total length of edges that need reinforcing, giving your answer in metres.

(2 marks)

Solution
$L = 4 \times (56 + 33 + 21) = 4 \times 110 = 440 \text{ cm}$ Length of edges is 4.4 metres.
Specific behaviours
✓ correct sum of edges ✓ correct length in metres

- (d) Determine the length of the longest thin wooden rod that can be placed in the box if

- (i) the entire length of the rod must be in contact with the base of the box. (2 marks)

Solution
Rod lies diagonally across base: $l = \sqrt{56^2 + 33^2}$ $= 65 \text{ cm}$
Specific behaviours
✓ indicates use of Pythagoras' Theorem with correct dimensions ✓ correct length

- (ii) the rod can lay in any position. (2 marks)

Solution
Rod lies from lower corner to top diagonally opposite corner: $l = \sqrt{65^2 + 21^2}$ $= 68.3 \text{ cm}$
Specific behaviours
✓ indicates use of Pythagoras' Theorem with correct dimensions ✓ correct length

Question 9

(7 marks)

The formula below calculates the monthly repayment P , in dollars, when a \$2500 loan is repaid over a period of n years and the interest rate on the loan is r percent per annum:

$$P = \frac{2500mi}{m-1} \text{ where } m = (1+i)^{12n} \text{ and } i = \frac{r}{1200}.$$

The formula for P is used to generate the values in the shaded cells from C3 onwards in the spreadsheet shown below.

	A	B	C	D	E	F	G
1			Repayment period in years (n)				
2			1	2	3	4	5
3	Interest	3	211.73	107.45	72.70	55.34	44.92
4	rate (r)	4	212.87	108.56	73.81	56.45	46.04
5		5	214.02	109.68	74.93	57.57	47.18

- (a) What is the monthly repayment when the loan is repaid over 2 years and the interest rate is 4% per annum. (1 mark)

Solution
Monthly repayment is \$108.56
Specific behaviours
✓ correct repayment

- (b) Determine the values of i and m used to calculate the value of P shown in cell G3 of the spreadsheet. (2 marks)

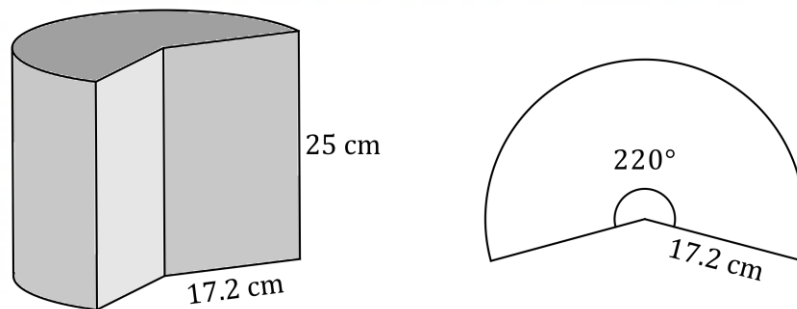
Solution
$r = 3, \quad i = \frac{3}{1200} = 0.0025$
$n = 5, \quad m = (1 + 0.0025)^{12 \times 5} = 1.1616$
Specific behaviours
✓ correct value of i
✓ correct value of m

- (c) Determine the value that would appear in cell H9 of the spreadsheet if the repayment periods in row 2 and interest rates in column B continue to increase as shown. (4 marks)

Solution
In cell H9, $r = 9$ and $n = 6$.
$i = 9 \div 1200 = 0.0075$
$m = (1 + 0.0075)^{12 \times 6} = 1.71255$
$P = \frac{2500 \times 1.71255 \times 0.0075}{1.71255 - 1} = 45.06$
Specific behaviours
✓ correct values for r and n
✓ calculates value of i
✓ calculates value of m
✓ correct value

Question 10 (Rounding Question)**(10 marks)**

The diagram below shows a prism of height 25 centimetres and detail of its cross section. The prisms are made of concrete and are used for garden edging.



The cross section is a sector of a circle of radius 17.2 centimetres with a central angle of 220° .

- (a) Show that the area of the cross section is 568 square centimetres, round the answer to the nearest square centimetres. (2 marks)

Solution
$A = \pi \times 17.2^2 \times \frac{220}{360}$ $= 567.97$ $\approx 568 \text{ cm}^2$
Specific behaviours
✓ shows correct calculation for area of whole circle ✓ shows multiplication by correct fraction

- (b) Determine the volume of the prism. Round the answer to the nearest cubic centimetre. (1 mark)

Solution
$V = 568 \times 25 = 14\,200 \text{ cm}^3$
Specific behaviours
✓ correct volume of prism (F/T)

- (c) Determine the length of the circular arc of the cross section, round the answer to the nearest centimetre. (2 marks)

Solution
$L = 2 \times \pi \times 17.2 \times \frac{220}{360}$ $= 66.0 \text{ cm}$
Specific behaviours
✓ correct expression for length ✓ correct length

- (d) Explain why the area of the curved surface of the prism is 1650 square centimetres. (2 marks)

Solution
If the curved surface was flattened it would be a rectangle of length equal to the height of the prism, 25 cm, and width equal to the length of arc of the sector, 66 cm. Hence $A = 25 \times 66 = 1650 \text{ cm}^2$.
Specific behaviours
✓ explains why area is rectangular (labelled graph is allowed) ✓ explains dimensions of rectangle

- (e) Determine the total surface area of the concrete prism. (3 marks)

Solution
Area of one rectangular face: $A = 25 \times 17.2 = 430$ Total surface area: $A = 2 \times 430 + 2 \times 568 + 1650$ $= 860 + 1136 + 1650$ $= 3646 \text{ cm}^2$
Specific behaviours
✓ calculates area of rectangular face ✓ shows calculation for total surface area using all five surfaces ✓ correct total surface area (F/T)

Question 11 (Money must be rounded to 2dp. If not, only take one mark off for the entire Q11) (12 marks)

Maureen has a special bank account that enables her to hold Australian dollars and several foreign currencies at the same time. She can exchange currencies at any time she wishes, and her account currently holds \$1150, €386 and ฿7540.

The buy and sell rates for one Australian dollar that her bank offers on these currencies are shown in the table below. The bank charges no conversion or other fees to account holders.

Currency (region, symbol)	Buy rate	Sell rate
Euro (European Union, €)	0.6044	0.5812
Baht (Thailand, ฿)	21.96	21.09

- (a) Explain why the buy rate is used when she converts Thai baht into Australian dollars.

(1 mark)

Solution
Because her bank is buying the currency from her.
Specific behaviours
✓ indicates bank is the buyer

- (b) Determine the balance of her account if she converted all her foreign currencies into Australian dollars.

(2 marks)

Solution
Euro: $€386 \div 0.6044 = \$638.65$
Baht: $฿7540 \div 21.96 = \$343.35$
Balance: $\$1150 + \$638.65 + \$343.35 = \2132.00
Specific behaviours
✓ correctly converts one currency
✓ correct account balance

Maureen wants to exchange some of her Australian dollars into euro so that her holding of this currency increases from €386 to €750.

- (c) Determine how many Australian dollars would remain in her account after the exchange is complete.

(3 marks)

Solution
Bank must sell her $€750 - €386 = €364$
This will cost $€364 \div 0.5812 = \$626.29$
Balance of AUD is $\$1150 - \$626.29 = \$523.71$.
Specific behaviours
✓ calculates currency required
✓ correct conversion rate and method
✓ correct account balance (F/T)

Maureen is a shareholder in the listed companies Marjoke and Nemesis that have current market prices of \$11.40 and \$1.73 respectively per share.

She owns 1860 shares in Marjoke that has recently declared a dividend of 26 cents per share, and 13 620 shares in Nemesis that has declared a dividend yield of 1.2%.

- (d) Determine the total dividend Maureen will receive. (3 marks)

Solution
Marjoke: $\$0.26 \times 1860 = \483.60
Nemesis: $0.012 \times \$1.73 \times 13\,620 = \282.75
Total dividend: $\$483.60 + \$282.75 = \$766.35$
Specific behaviours
✓ one correct dividend
✓ second correct dividend
✓ correct total dividend

- (e) Calculate the percentage dividend yield for shares in Marjoke. (2 marks)

Solution
Yield is $0.26 \div 11.40 = 0.0228$
Percentage yield is 2.28%
Specific behaviours
✓ indicates correct method
✓ correct percentage dividend yield

- (f) Determine the price-to-earnings ratio for Nemesis, given that it has annual earnings per share of 6.2 cents. (1 mark)

Solution
Price-to-earnings ratio is $173 \div 6.2 = 27.9$
Specific behaviours
✓ correct ratio

Question 12

(10 marks)

A small supermarket sells several varieties of canned fish. The spreadsheet extract below shows how much they pay their supplier for packs of cans and their selling price for individual cans. For example, they sell a 95 gram can of tuna for \$1.20 having paid their supplier \$27 for a pack of 36 cans which equates to a cost of 75 cents per can. All prices include GST at 10%.

Item	Supplier			Supermarket
	Pack cost (\$)	Cans per pack	Unit cost (\$)	Selling price (\$)
Tuna 95 g	27.00	36	0.75	1.20
Tuna 185 g	30.00	24	1.25	1.70
Tuna 425 g	44.10	12	2.45	3.30
Sardines 110 g	93.60	48	A	2.75
Mackerel 125 g	B	28	2.80	3.50
Salmon 100 g	29.70	C	1.65	2.45

- (a) Determine the values of A , B and C in the table above.

(3 marks)

Solution
$A = 93.60 \div 48 = 1.95$
$B = 28 \times 2.80 = 78.40$
$C = 29.7 \div 1.65 = 18$
Specific behaviours
✓ correct value of A
✓ correct value of B
✓ correct value of C

- (b) Determine the percentage profit the store makes when they sell a can of mackerel.

(2 marks)

Solution
Profit: $3.50 - 2.80 = 0.70$
Percentage profit: $0.7 \div 2.80 = 0.25 = 25\%$
Specific behaviours
✓ indicates correct profit
✓ calculates percentage profit

- (c) Calculate the amount of GST included in the selling price of a can of sardines.

(1 mark)

Solution
GST: $2.75 \div 110 \times 10 = \0.25
Specific behaviours
✓ correct amount

- (d) The small supermarket is about to discount their selling price of a 95 gram can of tuna by 30%.

- (i) Calculate the discounted price. (1 mark)

Solution
Discounted price: $1.20 \times 0.70 = \$0.84$
Specific behaviours
✓ correct price

- (ii) Use the unit cost method to determine which size of tuna offers the best and which offers the worst value for money for someone who shops at the supermarket while the discount is in place. (3 marks)

Solution
Unit cost for each size
Tuna 95 g: $84 \div 95 = 0.88$ cents per gram
Tuna 185 g: $170 \div 185 = 0.92$ cents per gram
Tuna 425 g: $330 \div 425 = 0.78$ cents per gram
Best value is the 425 gram size and worst is the 185 gram size.
Specific behaviours
✓ one correct unit price ✓ all correct unit prices ✓ correctly states best and worst sizes (Students may use different unit prices such as cents/100 gram or dollar/kg, no penalty given the calculation is correct.)

Question 13

(10 marks)

Angus is a single person who is the main carer for one child and qualifies for a maximum fortnightly government parenting payment of \$1007.50. However, this amount reduces by 40 cents for every dollar of income over \$220.60 that Angus earns in a fortnight.

Angus has a part-time job that pays him \$38.50 per hour and he works for 12 hours each week.

- (a) Determine his fortnightly income from this job. (2 marks)

Solution
Weekly income: $\$38.50 \times 12 = \462 .
Fortnightly income: $\$462 \times 2 = \924 .
Specific behaviours
✓ correct expression for weekly income
✓ correct fortnightly income

- (b) Determine the fortnightly sum of his income and reduced parenting payment. (3 marks)

Solution
Income over threshold: $\$924 - \$220.60 = \$703.40$.
Reduction in payment: $\$703.40 \times 0.4 = \281.36 .
Sum of income and payment is $\$924 + \$1007.50 - \$281.36 = \1650.14 .
Specific behaviours
✓ indicates income over threshold
✓ calculates reduced payment
✓ calculates sum

Betty works from home selling draw tickets for a charity and is the main carer for two children. She qualifies for a maximum fortnightly government parenting payment of \$712.30 and has a partner who is paid an annual salary of \$50 080. In each fortnight, the amount of her parenting payment reduces by 50 cents for every dollar of income she earns over \$150 and by 60 cents for every dollar that her partner earns over \$1368.

- (c) Determine her partners fortnightly pay F using the formula $F = S \div 313 \times 12$, where S is the annual salary. (1 mark)

Solution
$F = \$50\,080 \div 313 \times 12 = \1920 .
Specific behaviours
✓ correct pay

- (d) Determine the parenting payment that Betty would receive in a fortnight if she sold \$2560 worth of tickets in that time, from which she earns a commission of 12.5%. (4 marks)

Solution
Commission for Betty: $\$2560 \times 0.125 = \320 .
Reduction in payment for Betty: $(\$320 - \$150) \times 0.5 = \$85$.
Reduction in payment for partner: $(\$1920 - \$1368) \times 0.6 = \$331.20$.
Parenting payment: $\$712.30 - \$85.00 - \$331.20 = \296.10 .
Specific behaviours
✓ calculates commission
✓ correctly calculates one reduction
✓ correctly calculates second reduction
✓ correct payment (F/T)

Question 14

(9 marks)

- (a) Calculate the value of an investment of \$5800 after 90 days when it earns simple interest at a rate of 7.3% per annum. (3 marks)

Solution
$I = \$5800 \times \frac{7.3}{100} \times \frac{90}{365}$ $= \$104.40$ Future value of investment will be $\\$5800 + \\$104.40 = \\$5904.40$
Specific behaviours
✓ indicates correct use of simple interest formula ✓ correctly calculates interest ✓ correct value of investment

- (b) An investor is considering placing \$17 500 into an account that advertises compound interest at a rate of 5.52% per annum.

- (i) Determine how much interest the investment would earn in 3 years if the compounding period was annually. (3 marks)

Solution
$F = \$17\,500 \left(1 + \frac{5.52}{100}\right)^3$ $= \$20\,560.91$ Interest earned is $\\$20\,560.91 - \\$17\,500.00 = \\$3060.91$
Specific behaviours
✓ indicates correct use of compound interest formula ✓ correctly calculates future value ✓ correct interest

- (ii) Determine the value of the investment at the end of 2 years if the compounding period was monthly. (3 marks)

Solution
Number of compounding periods is $2 \times 12 = 24$. $F = \$17\,500 \left(1 + \frac{5.52}{12 \times 100}\right)^{24}$ $= \$17\,500(1.0046)^{24}$ $= \$19\,537.74$ Future value of investment is \$19 537.74
Specific behaviours
✓ indicates correct number of compounding periods ✓ indicates correct use of compound interest formula ✓ correctly calculates future value

Question 15**(10 marks)**

Five devices, numbered from 1 to 5, can all communicate with each other on a Wi-Fi network, but not always directly. The following list shows which devices each can communicate directly with:

- device 1 can communicate directly with device 4
- device 2 can communicate directly with device 5
- device 3 can communicate directly with device 4
- device 4 can communicate directly with devices 1, 3 and 5
- device 5 can communicate directly with devices 2 and 4

No device can communicate directly with itself.

- (a) Construct matrix A to represent this network, with element $a_{i,j}$ equal to 1 if device i can communicate directly with device j , and equal to 0 if not. (2 marks)

Solution	
$A =$	$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$
Specific behaviours	
✓ symmetrical matrix with zeros and ones	
✓ correct matrix	

- (b) What information does the sum of the elements in the second column of matrix A represent? (1 mark)

Solution
The number of devices that can communicate directly with device 2.
Specific behaviours
✓ correct description
Alternate: there is one way to directly communicate to device 2.

Let matrix $B = A^2$.

- (c) Determine matrix B . (2 marks)

Solution	
$B =$	$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 3 & 0 \\ 1 & 0 & 1 & 0 & 2 \end{bmatrix}$
Specific behaviours	
✓ any correct row or column	
✓ correct matrix	

- (d) State the value of element $b_{3,2}$ in matrix B and state what it represents. (2 marks)

Solution
$b_{3,2} = 0$
It is not possible for device 3 to communicate with device 2 via one other device.
Specific behaviours
✓ correct value ✓ correct description

The energy required to power devices 1 and 3 is 11 watts, whilst the other devices require 15 watts.

- (e) Represent the energy requirement of the five devices in row matrix E . (1 mark)

Solution
$E = [11 \quad 15 \quad 11 \quad 15 \quad 15]$
Specific behaviours
✓ correct matrix

Matrix E can be multiplied by matrix F to obtain matrix D , where $d_{1,1}$ represents the total energy required to power all the networked devices.

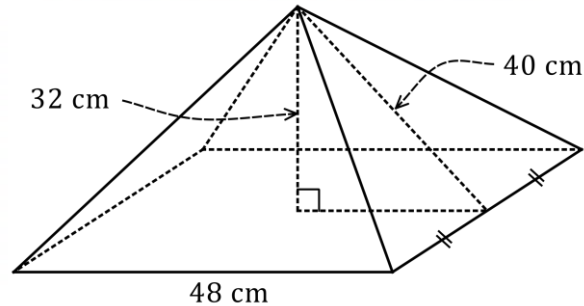
- (f) Write down matrix F and matrix D . (2 marks)

Solution
$F = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad D = [67]$
Specific behaviours
✓ one correct matrix ✓ second correct matrix

Question 16**(10 marks)**

The vertex of the right pyramid P shown below lies 32 centimetres directly above the centre of its square base and 40 centimetres from the midpoint of each side of its base.

The square base has sides of length 48 centimetres.



- (a) Determine the volume of the right pyramid P .

(2 marks)

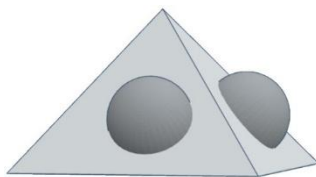
Solution
$V = \frac{1}{3} \times 48^2 \times 32$ $= 24\,576 \text{ cm}^3$
Specific behaviours
✓ substitutes into correct formula ✓ correct volume

- (b) Determine the total surface area of the right pyramid P .

(3 marks)

Solution
Square base: $A = 48^2 = 2304$
One triangular face: $A = \frac{1}{2} \times 48 \times 40 = 960$
Total surface area: $A = 2304 + 4 \times 960 = 2304 + 3740 = 6144 \text{ cm}^2$
Specific behaviours
✓ area of base ✓ area of one triangular face ✓ correct total surface area

The composite solid shown below is formed by gluing the circular face of a hemisphere of radius 9.9 centimetres to **each** triangular face of the right pyramid P .



- (c) Determine the volume of the composite solid.

(2 marks)

Solution
Volume of one hemisphere: $V = \frac{1}{2} \times \frac{4}{3} \times \pi \times 9.9^3 = 2032$ Volume of composite solid: $V = 24\,576 + 4 \times 2032 = 24\,576 + 8128 = 32\,704 \text{ cm}^3$
Specific behaviours
✓ calculates volume of one hemisphere ✓ correct composite volume

- (d) Determine the total surface area of the composite solid.

(3 marks)

Solution
Surface area of one hemisphere: $A = \frac{1}{2} \times 4\pi \times 9.9^2 = 616$ Area of circle lost: $A = \pi \times 9.9^2 = 308$ Total surface area of composite solid: $A = 6144 + 4 \times 616 - 4 \times 308$ $= 6144 + 2464 - 1232$ $= 7376 \text{ cm}^2$
Specific behaviours
✓ calculates surface area of one hemisphere ✓ calculates area of one circle ✓ correct total surface area

Question 17

(10 marks)

- (a) A solid cone made from wood has a radius of 5.7 centimetres and a perpendicular height of 17.6 centimetres.

- (i) Determine the slant height of the cone.

(2 marks)

Solution
$s = \sqrt{5.7^2 + 17.6^2}$ $= 18.5 \text{ cm}$
Specific behaviours
✓ indicates correct use of Pythagoras' Theorem ✓ correct slant height

- (ii) Determine the total surface area of the cone.

(2 marks)

Solution
$A = \pi(5.7)(18.5) + \pi(5.7)^2$ $= 331 + 102$ $= 433 \text{ cm}^2$
Specific behaviours
✓ correct area of curved side or base seen ✓ correct total surface area (F/T)

- (b) The volume of a cylinder of radius 4.5 centimetres is 250 cubic centimetres. Determine the volume of a similar cylinder whose radius is 15.3 centimetres. (3 marks)

Solution
Length scale factor is $15.3 \div 4.5 = 3.4$
Volume scale factor is $3.4^3 = 39.304$
Volume of similar cylinder is $250 \times 39.304 = 9826 \text{ cm}^3$
Specific behaviours
✓ calculates length scale factor ✓ obtains volume scale factor ✓ correct volume

- (c) The surface area of a prism of height 5.5 centimetres is 45 square centimetres. Determine the height of a similar prism whose surface area is 352.8 square centimetres. (3 marks)

Solution
Area scale factor is $352.8 \div 45 = 7.84$
Length scale factor is $\sqrt{7.84} = 2.8$
Height of similar prism is $5.5 \times 2.8 = 15.4 \text{ cm}$.
Specific behaviours
✓ calculates area scale factor ✓ obtains length scale factor ✓ correct height (F/T)

Supplementary page

Question number: _____

